

Autodesk® Scaleform®

Scaleform Best Practices Guide

Scaleform 4.0

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: 3.0
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はじめに

Autodesk® Scaleform® 4.0

Adobe Flash®

Flash

ActionScript™ (AS)

(HUD)

HUD

HUD

Scaleform 3.0

Flash

Web

Scaleform

Flash AS

(

PC

)

(

PC

)

(UI)

UI

Flash

FLA

Flash Studio CS5

XML
.FLA

1

Flash

AS

AS

Flash

FLA

Flash

:

• Flash

AS

:

。

- FLA AS :
- AS

- :
-

- Flash AS :
- Flash

Flash

Flash AS

Flash :

-
-

ActionScript

1

Flash

Flash

一般的な使用法

- 1
- UI (Flash [])

Scaleform 3.3
Scaleform Common Lightweight Interface Kit (CLIK™)

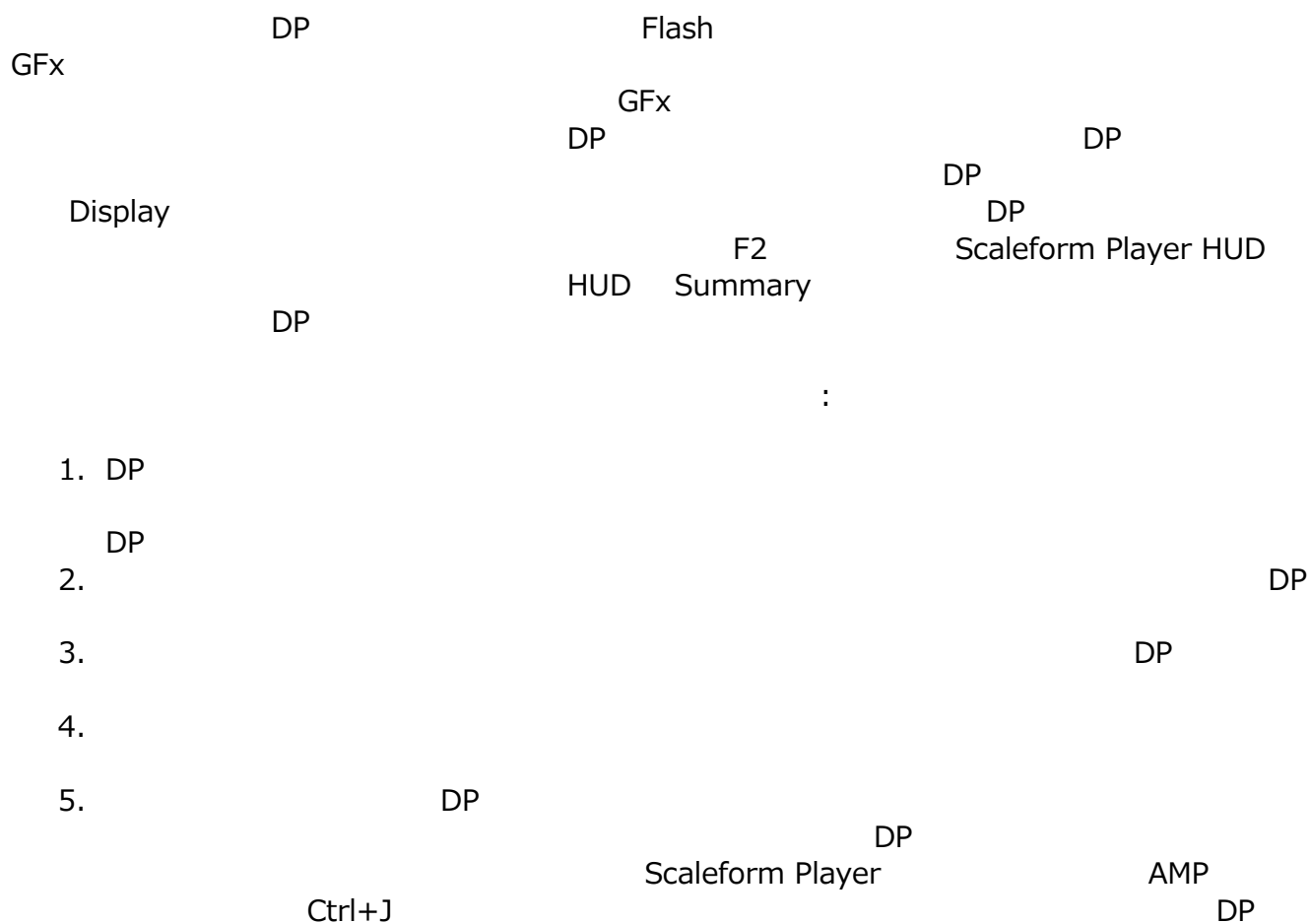
セットアップの検討事項

-
- - Flash
 - Flash
 - AS
 - AS2
 - AS3
 - MMORPG
 - HUD
 - SWF
 - API
 - C++ API (Gfx::Movie
 - Scaleform 3.3
 - Flash
 - CPU
 - CD
 - DVD
- SWF
- - UI
 - :
 - HUD : 400K-1.5M
 - / : 800K-4M
 - ActionScript ():
 - 700K-1.5M
 - : 2M-10M+

コンテンツ作成時のメモリとパフォーマンスについての考察

Flash

描画プリミティブ



ムービー クリップ°

1. Advance

2.

3. AS2) and `visible=false` rather than `alpha=0` (in AS3) `_alpha=0` `_visible=false`(in "stop()")

アートワーク

ビットマップ グラフィック ベクター グラフィック

Flash

Scaleform

ベクター グラフィック

()
()

●

●

○

Flash

•

○

○

•

O

○

●

•

○

O

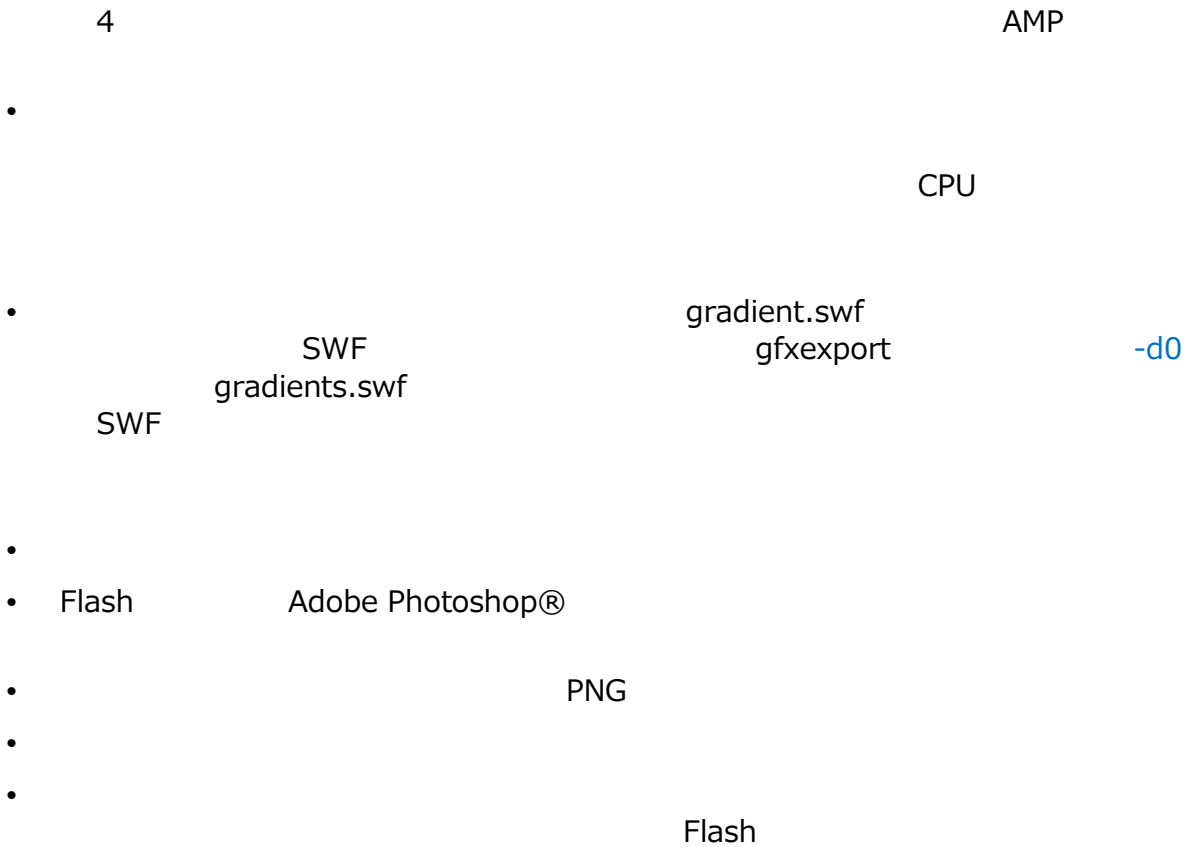
◦		2
• Flash API	API	1
•		
	Flash Studio	
	Scaleform	EdgeAA
•		
•		
• EdgeAA		1
/	/	

ビットマップ

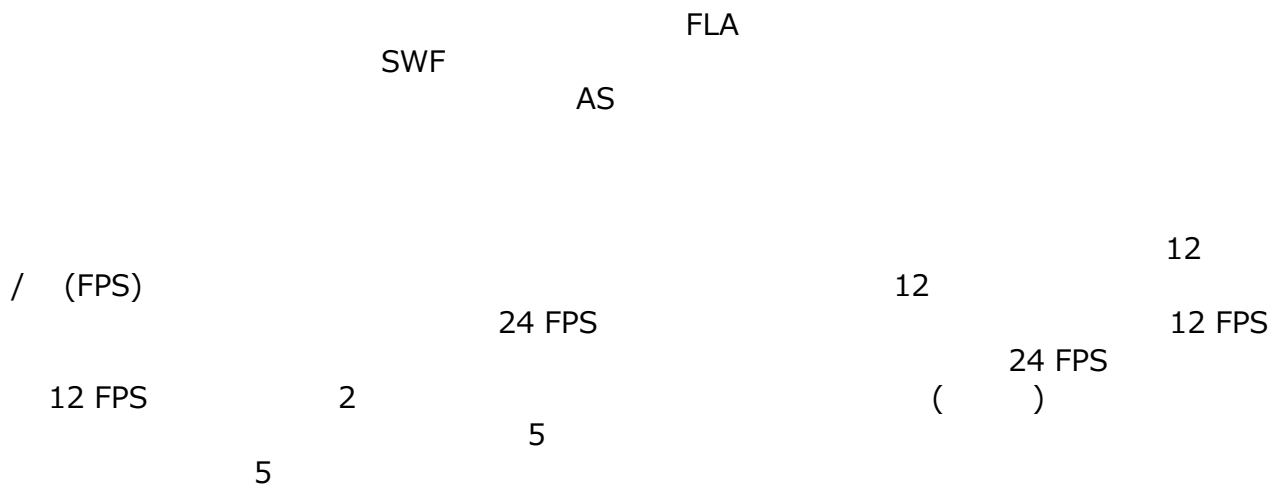
Flash 10

SWF `_alpha`(in AS2) or `alpha` (in AS3) 0
 `_visible`(in AS2) or `visible` (in AS3)
 false Scaleform `_alpha/alpha` 0 CPU
 CPU ActionScript false SWF
 `_visible/visible` false
 :

- 2 /
 16x32 256x128 1024x1024 512x32
- JPEG
- gfxexport
 DXT SWF GFx
 DXT gfxexport
 `-qp` `-qh` DDS ()
- Edge AA
- ActionScript /
- SWF/GFX . SWF SWF
- 1024x1024 JPEG
 4 MB
- UI 4 () gfxexport `-i DDS`



アニメーション



onEnterFrame(in AS2) or Event.ENTER_FRAME (in AS3)

onEnterFrame/Event.ENTER_FRAME

setInterval

onEnterFrame/EVENT.ENTER_FRAME

setInterval

30 40 FPS
CPU

Flash

UI

:

•

•

/

/
_visible/visible = true/false

•

◦

1

◦

Flash

◦ Tween
Scaleform CLIK Tween (gfx.motion.Tween in AS2 or
scaleform.clik.motion.Tween in AS3)
Tween

Flash

•

60

Flash

60 FPS

Flash

•

•

•

(

)

•

-

-

CPU

(
()

CPU
)

4

/

-

()

テキストとフォント

-

gfxexport (48)

SlotHeight

DP
)

(

DP

-

DP 1

-

1

30 FPS

-

("TextField.variable")

-

"htmlText"

HTML

-

gfxexport -fc -fcl -fcm

(

)

"Font and Text Configuration Overview"

- fontlib
("Font and Text Configuration Overview"
)
- TextField 1 1
1 DP
- 1 (1
)
- ()
-
- DrawText API DrawText API
Flash Scaleform C++
Flash UI DrawText API C++ 1

の最適化

ActionScript

AS

AS

一般的な *ActionScript* のガイドライン

AS

- SWF Flash 8
- AS

AS

[attachMovie](#)

FLA

- AS
-

•

- “if”

- [on\(\)](#) [onClipEvent\(\)](#) [onEnterFrame](#)
[onPress](#)

- AS

Flash AS

([onClipEvent](#) [on](#))

([gotoAndPlay](#) [play](#) [stop](#))

- , [gotoAndPlay/gotoAndStop](#) ,

- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- Gfx::Movie::SetVariableArray
 1
 1
 C++ C++ AS 1
 1
- 1
 Scaleform
 Scaleform
- Object
 ◦ Object
- eval() 1
- myArr.length : Array.length
- ```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
 trace(fontArr[i]);
}
```

```

:
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
 trace(fontArr[i]);
}

```

- ( null )
- removeListener()
- - AS VM
  - com.xxx.yyy.aaa.bbb VM if AS
  - aaa bbb com xxx yyy if
  - com xxx xxx yyy
  - Flash SWF
  - c58923409876.functionName()
- - AS SWF SWF
- AS
- - SWF [ ] trace()
  - [Flash] [Trace
  - trace
- 
- - 1 SWF AS ( : foo.bar.CustomClass)
  - SWF SWF
  - SWF

delete

:

```
delete foo.bar.CustomClass
```

- 100%  
( )
- onEnterFrames
- - ( ) Scaleform

## ループ

- 
- 
- 
- -

## 関数

- 
- with

## 変数 プロパティ

- 
- "var"  
ActionScript "var"
-

- 
- null
- SWF  
null
- AS getter setter

## ActionScript 3 の汎用ガイドライン

AS3

### データ タイプの厳格な指定

- 
- Object Object AS3
- Array Vector  
Array 4294967295  
Array
- Vector Vector<\*>
- Vector<\*> Vector
- Vector<String> Vector<int>  
Vector<YourFavoriteClass> Vector<\*>
- Object  
flash.utils.Dictionary
- GfX Flash
- AS2 AS3 AS3

## Advance

Advance

6 :

1. AS

onEnterFrame/Event.ENTER\_FRAME

2.

Invoke

UI

3.

(AS3)

true

(\_visible(AS2) or visible  
stop())

)

Advance

1 1

どれも

4. \_global.noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance  
(AS3)

Advance

"true"

Advance

)

(  
Flash

Flash

(

)

\_global.gfxExtensions(AS2) or

scaleform.gfx.Extensions.enabled (AS3) true

Scaleform

5.

Advance

6.

## onEnterFrame and Event.ENTER\_FRAME

onEnterFrame(AS2) or Event.ENTER\_FRAME (AS3)

onEnterFrame/Event.ENTER\_FRAME

setInterval setTimeout

setInterval

:

- clearInterval
- setInterval    setTimeout    onEnterFrame/Event.ENTER\_FRAME  
onEnterFrame/Event.ENTER\_FRAME

onEnterFrame                      delete                      :

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

onEnterFrame    null    undefined                      (    this.onEnterFrame = null;)  
onEnterFrame                      onEnterFrame  
Scaleform

## ***ActionScript 2 の最適化***

### イベントと イベント

onClipEvent()    on()                      onEnterFrame    onPress  
:

- 
- onClipEvent    on  
this    \_global    arguments    super  
256

onLoad

onEnterFrame                      :                      onClipEvent(construct)

```
onClipEvent(construct)
{
 this.onLoad = function()
 {
 //
 }
}
```

onClipEvent(load)

## キーワード

var

AS

var

AS

2

:

```
var i = 1000;
countIt = function()
{
 num = 0;
 for(j=0; j<i; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}
```

:

```
var i = 1000;
countIt = function()
{
 var num = 0;
 var ii = i;
 for(var j=0; j<ii; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}
```

## プリキャッシュ

) (var

```
function foo(var obj:Object)
{
 for (var i = 0; i < obj.size; i++)
 {
 obj.value[i] = obj.num1 * obj.num2;
 }
}
```

```
function foo(var obj:Object)
{
 var sz = obj.size;
 var n1 = obj.num1;
 var n2 = obj.num1;
 for (var i = 0; i < sz; i++)
 {
 obj.value[i] = n1*n2;
 }
}
```

```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
 // ;
}
```

## 長いパスのプリキャッシュ

:

```
mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;
```

:

```
var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;
```

## 複雑な式

C

:

```
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;
```

:

```
for(i=0; i<3; i++)
{
 this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
```

```
0x0000000;
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 _splqueue[slot_i].gosplash.text = _splstring;
 _splqueue[slot_i].gosplash2.text = splstring;
 _splqueue[slot_i].gosplash2.textColor = 0x0000000;
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 var elem = _splqueue[slot_i];
 elem.gosplash.text = _splstring;
 var gspl2 = elem.gosplash2;
 gspl2.text = splstring;
 gspl2.textColor = 0x0000000;
}
```

## の開発

(HUD)

Scaleform HUD

HUD

## 4.1

HUD

HUD

HUD

# C++

## 複数の SWF ムービー ビュー

HUD

80K

HUD

SWF

■

■

1. 別々のスレッド上で *Advance* を呼び出す機能。

## Flash

HUD

## Advance

( )

AS

Flash

## Advance

## Flash

## Advance

HUD

## Advance

## Advance

HUD

## Advance

: Scaleform

Gfx::Movie

## Advance

|  |        |                    |       |
|--|--------|--------------------|-------|
|  | Invoke | Display<br>Advance | Input |
|--|--------|--------------------|-------|

2. *render-to-texture* キャッシングを利用する。 HUD

HUD

## 複数の SWF を含む 1 つのムービー ビュー

|       |                              |   |
|-------|------------------------------|---|
| Flash | AS <a href="#">loadMovie</a> | 1 |
|-------|------------------------------|---|

Display

|       |   |
|-------|---|
| Flash | 1 |
|-------|---|

1. [\\_visible = false](#)(AS2) or  
[\\_global.noInvisibleAdvance](#)(AS2) or  
[scaleform.gfx.Extensions.noInvisibleAdvance](#) (AS3) true Advance  
作成するときに可能な最も重要な事項の つ

は :

|                                                        |                                                             |
|--------------------------------------------------------|-------------------------------------------------------------|
| <a href="#">_visible/visible = false</a>               |                                                             |
| ( :<br>Advance<br>Flash<br>HUD<br>)                    | なく                                                          |
| / HUD                                                  | (                                                           |
| Advance<br>scaleform.gfx.Extensions.enabled (AS3) true | <a href="#">_global.gfxExtensions</a> (AS2) or<br>Scaleform |

2. [SetVariableArray](#) 1 Invoke

|   |   |   |
|---|---|---|
| ( | ) | 1 |
|---|---|---|

|                                  |          |
|----------------------------------|----------|
| <a href="#">SetVariableArray</a> | (        |
| )                                | 1 Invoke |

1

3. HUD onEnterFrame(AS2) or  
Event.ENTER\_FRAME (AS3)  
HUD onEnterFrame/Event.ENTER\_FRAME  
Advance  
onEnterFrame
4. AS  

tick  
AS Invoke

Flash

30-40 FPS

15-20 FPS

HUD
- 5.

## 1つのムービー ビュー

- Flash

C++  
Flash

( )

Flash
- :

4.2
1. 

Flash

Scaleform

HUD

Flash
  2. 

C++  
C++  
Render::Renderer  
C++  
AS

Flash

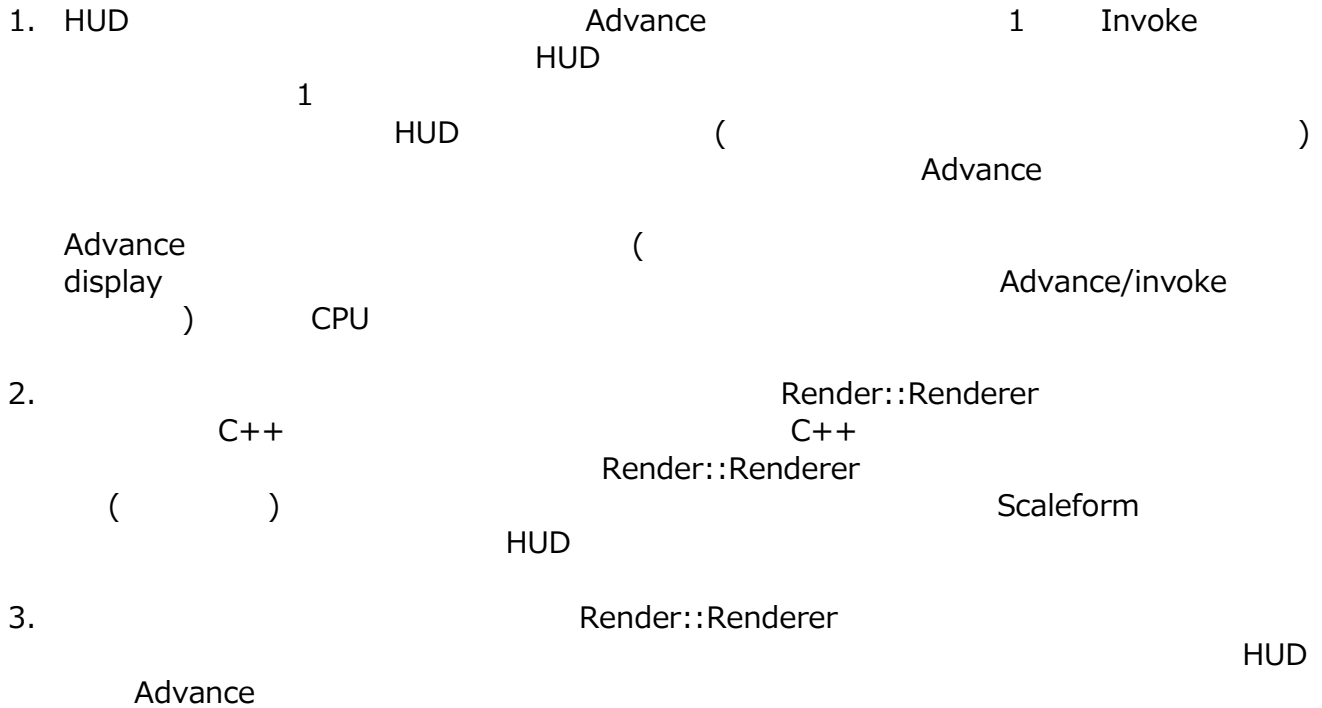
Flash  
RenderString

1

## 1つのムービー ビュー (上級編)

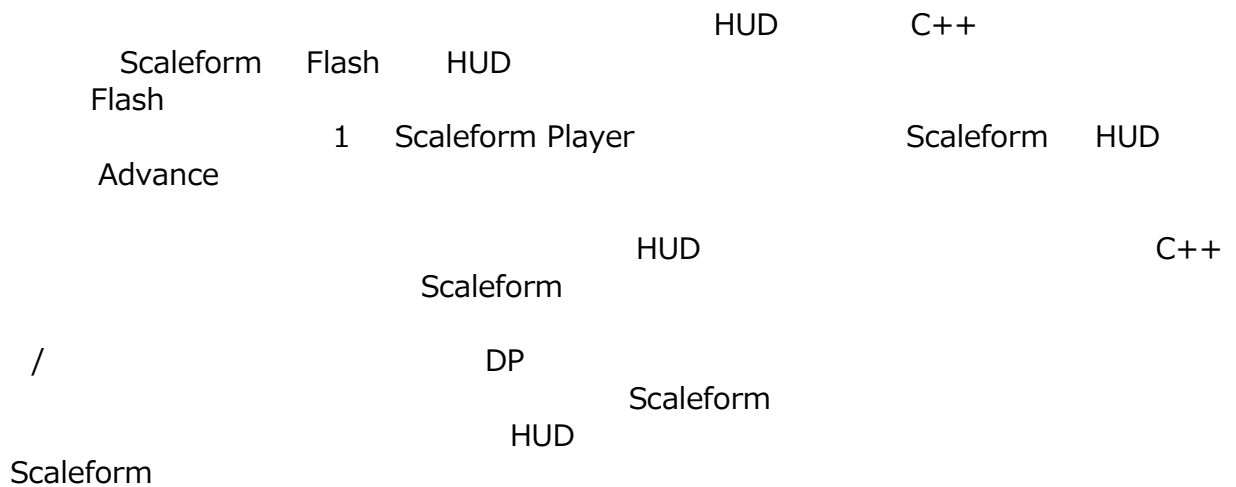
HUD  
4.3

:



: Scaleform -Unreal® Engine 3

## **Flash を使用しないカスタムの HUD 作成**



[Scaleform](#) / [Flash](#)  
[Scaleform](#) C++  
[DrawText API 1](#) [Scaleform 2](#) [HUD 1](#)  
[Scaleform](#) / [Scaleform FAQ](#) ["Font & Text"](#) [Font and](#)  
[Text Configuration Overview](#) / [DrawText API](#)

- HUD :
- - ["Graphics Rendering & Special Effects"](#) 1 2 FAQ
  - PC Wii™ (
    - Gfx::Movie::SetMouseCursorCount(0);
    -
  - / HUD  
noInvisibleAdvance (AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3)  
[\\_visible\(AS2\) or visible \(AS3\) = false](#)
  - Invoke 1 2 Invoke  
( )  
( )
  - [2.1](#)  
FAQ ["Art & Assets"](#)

## 一般的な最適化のヒント

### Flash のタイムライン

- Flash 2 FLA
- FPS
- 
- 
- 
- AS /
- ([ 1] [ 2] )

### 全般的なパフォーマンスの最適化

- 3 1
- 
- `trace()`
- ( AS )
- AS actions
- AS 1

## その他の資料

ActionScript 2.0 Best Practices

[http://www.adobe.com/devnet/flash/articles/as\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/as_bestpractices.html)

Flash 8 Best Practices

[http://www.adobe.com/devnet/flash/articles/flash8\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html)

Flash ActionScript 2.0 Learning Guide

[http://www.adobe.com/devnet/flash/articles/actionscript\\_guide.html](http://www.adobe.com/devnet/flash/articles/actionscript_guide.html)